Accreditation and Quality Assurance (2025) 30:313–323
https://doi.org/10.1007/s00769-025-01643-z

RESEARCH

Determination of chlorpyrifos using GC-FID in pesticides and human plasma; method validation with greenness assessment

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Received: 19 January 2025 / Accepted: 3 May 2025 / Published online: 31 May 2025
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Abstract

About 3.70 million tons of pesticides are consumed annually to improve agricultural revenues. Organophosphate (OP) is a significant pesticide class, representing approximately 40% of pesticides' global sales. Chlorpyrifos (CPF) has been an OP pesticide commonly used to prevent horticultural and domestic pests for the past 25 years. Although CPF was banned from household products by the United States Environmental Protection Agency (US-EPA) in the year 2000, its current sales in the US alone exceed 5,000 tons. Other authorities, such as European, Canadian and Egyptian, have also discontinued its use since 2020. Therefore, it is necessary to analyze CPF as a counterfeit pesticide in products and biological samples in the coming years. CPF has high toxicity, representing neurological and cardiovascular threats. The proposed research aims to develop a novel, fast, and cost-effective methodology to estimate CPF in pesticide products and biological human plasma using gas chromatography (GC) with a flame ionization detector (FID). Notably, only a few GC-FID methods were reported for CPF analysis in marketed products. Still, none has yet been reported for its determination in plasma using GC-FID despite being greener, cheaper, and simpler than GC-tandem mass spectrometry (MS/MS). Both procedures were validated according to the International Council for Harmonization (ICH) guidelines. The retention time was 13.3 min. Linearity was perceived in the concentration of 20.0–100.0 µg/mL, with a coefficient of determination of 0.9909 for analytical and 0.9905 for bio-analytical application, respectively. The detection (LOD) and quantification limits (LOQ) were 0.29 and 0.87 µg/mL. Further, the greenness of the developed technique was assessed and compared to some other reported methodologies. The validation and greenness results indicate that both methods are reliable and sensitive, making them suitable for routine quality control of CPF in marketed products and biological samples.

Keywords GC-FID · Chlorpyrifos · Bio-analytical method · Pesticide control · Human plasma

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Introduction

The world population is expected to reach 10 billion in 2050 [1]. The current significant growth in population requires a rise in agricultural and food production. This creates a strong demand for pesticides globally, with around 3.70 million tons used annually [2]. The countries that use the most pesticides include Brazil, the United States (US), Indonesia, Argentina, and China [3]. According to the World Health Organization (WHO), pesticides are chemical substances used to eliminate pests, such as insects, rodents, fungi, and unwanted plants (weeds) [4]. Moreover, pesticides have been categorized into different types, such as biodegradable and chemical pesticides. Chemical pesticides encompass first-generation inorganic compounds such as sulfur and arsenic and second-generation synthetic compounds, including organochlorines, organophosphorus, carbamates, and pyrethroids, which have various mechanisms of action [5–7]. Organophosphate (OP) pesticides are described as one of the most famous and widely used pesticides, whereas 40% of all commercial pesticides purchased from the market belong to this class because of their effectiveness in killing large numbers of pests [8].

OP compounds were initially formulated during the 1930s-1940s as human nerve gas agents, and most were later updated at lower doses as pesticides [9]. Since OP pesticides disperse readily in the ecological system, they are better tolerated than persistent organochlorine pesticides such as dichloro-diphenyl-trichloroethane (DDT), aldrin, and dieldrin. The United States Environmental Protection Agency (US-EPA) identifies 40 organophosphate insecticides, including commonly used varieties [10]. Organophosphorus compounds are extensively used for cultivation, farming, management of rodents, pest control, polymer processing, counterinsurgency, and household purposes [11, 12]. In 2010, the total usage of OPs reached approximately 300,000 tons, accounting for about 72% of China's total pesticide consumption [13]. OP pesticides should be sprayed within their optimum concentration required for agriculture to obtain the highest benefits for crops with human safety. For instance, phosalone and chlorpyrifos should be sprayed at 0.004 and 0.001-0.003 mg/kg(bw)/day, respectively [5]. However, the US-EPA declared OP pesticides as class I to IV for inhalation or oral poisoning [14]. Phosphorylation of the serine hydroxyl group located at the binding center of carboxyl ester hydrolases, particularly acetylcholinesterase (AChE), is the predominant biological mechanism of OP pesticides. Atropine competes with acetylcholine at muscarinic receptors and is the definitive cure for organophosphate poisoning. However, since it only acts on muscarinic receptors, Pralidoxime should also be administered to affect the nicotinic receptors [15]. According to the World Health Organization (WHO), poisoning from OP compounds is a global epidemic. It is estimated that there are three million cases worldwide each year, which includes one million accidental poisonings and two million cases of suicide attempts [16]. The number of annual human deaths from WHO-approved OP pesticides is about 300,000 deaths [17].

Among OP pesticides, Chlorpyrifos (CPF) [O, O-diethyl O-(3, 5, 6-trichlor-2-pyridyl) phosphorothioate] has been a commonly used wide-ranging insecticide since 1965 for the management of agricultural and domestic pests [18]. Chemical structure and properties are shown in Figure 1 and Table 1 [19], respectively.

It undergoes rapid hydrolysis to form primary metabolites and other intermediary by-products such as Chlorpyrifos-oxon, diethyl thiophosphate (DETP), desethyl-chlorpyrifos, and 3,5,6-trichloro-2 methoxy pyrimidine [20]. Besides its affordable price, it effectively eradicates numerous pests and has applications as an acaricide, insecticide and termiticide [21]. Despite that, tests have consistently reported significant health consequences from its usage, including disrupted brain growth in infants and learning difficulties similar to those caused by Alzheimer's disease in rats, both in cases of acute and prolonged administration [22, 23]. The principal adverse health effect is due to the inactivation of AChE found in most animals' nervous systems. Although organochlorine pesticides, such as DDT, were highly effective and replaced toxic lead arsenate, they have since been substituted with synthetic pyrethroids and organophosphorus pesticides, including CPF. This change occurred due to their persistence and bioaccumulation properties, which led to an international ban. Recently, many authorities have raised concerns about CPF, which may lead it to fall within the same trend known as "regrettable substitution" in the coming years [7, 24].

In recent times, particularly since 2020, several authorities have decided to withdraw and discontinue the use and

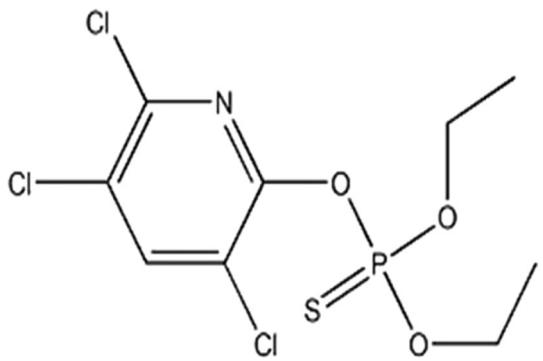


Fig. 1 Chemical structure of Chlorpyrifos

Table 1 Properties of Chlorpyrifos

Chemical name/ IUPAC name	[O, O-diethyl O-(3, 5, 6-trichlor-2-pyridyl) phosphorothioate]
Synonyms	Chlorpyrifos, Dursban, Lorsban
Chemical safety	Acute Toxicity
Molecular weight	350.6 g/mol
Water Solubility	Poor (0.73 mg/L at 20 °C) [19]
Physical appearance	White crystalline or irregularly flaked solid
Category	Pesticide

sale of CPF in the market. These authorities include the European Commission, Canada, Egypt, Vietnam, Saudi Arabia, and Indonesia. Furthermore, the European Commission has lowered the maximum residue levels of CPF to 0.01 mg/kg for all food and feed across Europe [25]. In the US, CPF should have been banned in 2001, but it was still being used, and that issue was discussed during this period [26, 27]. Therefore, it will be necessary to analyze CPF as a counterfeit in pesticide products in the upcoming years and in biological samples such as urine, blood, stomach, and liver tissues to determine exposure and for postmortem studies in the event of ingestion [28].

Analytical methods focused on using sensors [29, 30], spectroscopy [31, 32], and chromatography [26, 27] with different techniques were available to estimate CPF in various forms of farm products, soil, biological samples, and water. Gas chromatography (GC) is the most commonly used, and it can extract and analyze gaseous samples, liquid solutions, and volatile solids. Upon screening GC methods, shown in Supplementary materials Figure S1 and Table S2, CPF determination in pesticide products using GC-flame ionization detector (FID) was little. Furthermore, GC-FID was not used to determine CPF in plasma despite being greener, cheaper, and simpler than GC-mass spectrometry (MS). Concerning the foregoing, the current work focuses on developing, validating and bio-validating a green GC-FID method to determine CPF in pesticides and human plasma with the fundamental purpose of being simple, derivatization-free, efficient, and economically applied regularly.

Materials and methods

Materials

The Standard CPF (purity ≥98.6%) was purchased from Sigma-Aldrich (Missouri, United States). Lantrek®, a 20.0% CPF emulsifiable concentrate pesticide product (Chemi Enterprises, India), was procured from the Indian market. HPLC grades of acetonitrile (ACN) and methanol (MeOH) were obtained from Merck (New Jersey, United States). Plasma was obtained from different Indian blood banks and frozen at −20.0 °C until use.

Instrumentation

Chromatographic experiments were performed using Shimadzu's GC-2014 model with a flame ionization detector (FID). Data were collected and evaluated using GC solution software. Chromatographic separation was performed on the Zebron DB column (30 m x 0.25 mm x 0.50 µm), which can be operated within a temperature range of 40–320 °C (Max. 340 °C).

Chromatographic conditions

GC was operated using N_2 gas, at 100 kPa, as the carrier gas. Hydrogen and zero air were used as supporting gases in the FID for heat generation. The flow rate was maintained at 35 mL/min, and the column temperature was set at 80 °C for 1 minute, followed by a gradual rise to 280 °C at a rate of 15 °C/min. The septum injector and FID temperatures were 200 °C and 310 °C, respectively. For sample analyses, 1 µL was injected splitlessly.

Preparation of standard solutions

5.0 mg of CPF was weighed and dissolved in 5.0 mL ACN to prepare 1.0 mg/mL stock standard solution, which was stored in a refrigerator (2-8 °C). The stock solution prepared 5 concentration levels of 20.0, 40.0, 60.0, 80.0, and 100.0 µg/mL CPF dilutions for linearity.

Preparation of human plasma standards and QC samples

The CPF standard solution was spiked in 100.0 µL of blank human plasma to prepare the plasma samples at concentrations ranging from 20.0 to 100.0 µg/mL. After that, samples were mixed with 100.0 µL MeOH, used as a protein-precipitating agent, into an Eppendorf and whirled for 20 seconds by vortex mixer. Methanol was added again to make up the

volume to 1.50 mL. After that, the mixture was centrifuged at 10000 RPM for 10 minutes at 4 °C. The clear supernatant solution was filtered through a 0.20 µm syringe filter before injection into the GC-FID.

A blank plasma sample was prepared using the same procedure. Quality control samples (QCs) were prepared from separate stock solutions to avoid biased estimations by freshly spiking plasma with five concentration levels: lower limit of quantification (LLOQ) (20.0 µg/mL), lower quality control (LQC) concentration (40.0 µg/mL), middle quality control (MQC) concentration (60.0 µg/mL), higher quality control (HQC) concentration (80.0 µg/mL), and upper limit of quantification (UQC) (100.0 µg/mL).

Preparation of marketed products.

To prepare the marketed product sample, 2.50 µL of Lantrek® was dissolved in 10.0 mL ACN to prepare a 50.0 µg/mL concentration, which was then filtered using a 0.45µm syringe filter.

Analytical method validation

The suggested analytical methodology was tested for its system suitability, linearity, range, accuracy, precision, LOD, LOQ, Robustness, and Ruggedness per the International Council for Harmonization (ICH) Q2 (R1) guidelines [33].

System suitability, linearity, and range

1µL splitless injection of the standards was injected six times to assess system suitability parameters, including tailing factor and N. The linearity studies were performed by injecting the standards. The calibration curve was plotted using five concentrations, ranging from 20.0 to 100.0 µg/mL. These concentrations were 20.0, 40.0, 60.0, 80.0, and 100.0 µg/mL.

Accuracy, precision, and uncertainty

For accuracy, the market product (Lantrek®) was prepared at a concentration of 20.0 µg/mL and then spiked with three standards: 20.0, 40.0, and 80.0 µg/mL. The mean recovery percentages of three replicates of final concentrations 40.0,60.0 and 100.0 µg/mL were calculated.

The three standards, 20.0, 60.0, and 100.0 µg/mL, were injected and analyzed three times each under controlled laboratory conditions to evaluate repeatability and inter-day precision (over two different days), expressed as a relative standard deviation (RSD) percentage. The uncertainty of the analytical method at 95% confidence intervals (CI) was evaluated according to Leyva-Morales JB et al. [34] and Bastidas-Bastidaset al. [35], where the RSD percentage was calculated for accuracy results and subsequently multiplied by the coverage factor (for 95% = 2).

Limit Of detection (LOD) and limit of quantification (LOQ)

The Signal-to-Noise ratio was used to calculate the detection limit (LOD) and quantitative limit (LOQ), with typical ratios of 3:1 and 10:1, respectively.

Robustness and ruggedness

The 60 µg/mL standard was injected to ensure robustness while raising and lowering the injectors and detectors temperatures. The injector temperatures were 195 °C and 205 °C, while the detector temperatures were 308 °C and 312 °C. Additionally, for ruggedness, the analyst was changed in two days.

Bioanalytical method validation

The proposed bioanalytical methodology is validated according to the ICH M10 guidelines for bioanalytical method validation and study sample analysis [36].

Linearity, range, and selectivity

The same range of the analytical method was applied by spiking human plasma, and a calibration curve was constructed. Six plasma batches obtained from different sources were used for the selectivity studies.

Accuracy, precision, uncertainty, and recovery (extraction efficiency)

The bioanalytical method's accuracy and precision were evaluated by injecting LQC, MQC, and HQC as three replicates on two separate days for three weeks, resulting in six replicates for each QC sample each week under controlled laboratory conditions. The uncertainty of the bioanalytical method was determined using the same analytical validation approach.

Recovery studies were conducted using LLOQ, LQC, MQC, HQC, and UQC to assess the extraction efficiency. These QCs were extracted through the method's sample processing and extraction steps, and extraction efficiency was calculated as the percentage of the Bioanalytical Peak area to the Analytical Peak Area.

Sensitivity, matrix effect, and carryover

The extracted LLOQ was injected six times, and the %RSD was calculated to check sensitivity.

The matrix effect was evaluated by injecting three replicates of CPF spiked in 6 different plasma batches at LQC and HQC concentrations. The percentage recovery results of CPF from the spiked samples were procured on two separate days to study the effect of plasma matrices on the analytical results. During validation, blank samples were injected after HQC runs to assess carry-over.

Stability studies

LQC and HQC in plasma were stored at ambient temperatures and analyzed after 4, 8, 16, and 24 hours to assess Bench-Top (short-term) stability. Freeze–thaw stability was evaluated after subjecting QC samples to three cycles of freezing at − 20 °C and thawing at room temperature; there was a 12-hour interval between each cycle. For long-term stability, the samples were stored for ten days at − 20 °C. The average recovery percentages from three replicates were calculated.

Results and discussion

Optimization

The optimization step in GC analysis is of paramount importance for achieving accurate, precise, and efficient results. It enables the separation of analytes, enhances sensitivity, ensures reproducibility, facilitates method development and improves time and cost efficiency. By investing time and effort into optimization, analysts can optimize their GC methods to meet specific analytical goals and obtain reliable and meaningful data.

A comprehensive study investigated the different factors that can impact the GC system. The critical parameters, such as the column, run time, resolution, flow rate, and mobile phase, were optimized based on peak parameters and device suitability following ICH guidelines. In addition, the solubility of Chlorpyrifos was evaluated in various solvents. After careful evaluation, acetonitrile was the best solvent, with nitrogen as carrier gas at a flow rate of 35 mL/min, resulting in a resolution of more than 1.5 and a retention time of 13.3 minutes.

Experiments were also conducted to extract CPF from spiked human plasma using methanol, acetonitrile, and ethyl acetate. A mixture of plasma and CPF was used for protein precipitation studies. The experimental trial results showed that acetonitrile and ethyl acetate had lower recovery rates and peak responses (67.3 to 70.5%). Methanol on the other hand gave better peaks and recovery as would be shown later in bioanalytical method validation results. Therefore, methanol was chosen as the protein-precipitating agent. To perform the extraction, 100 µL of human plasma and 100 µL of CPF standard are added to an Eppendorf tube. After that, the mixture is vortexed for 20 seconds, followed by adding methanol to make a total volume of 1.5 mL. Next, the mixture is centrifuged for 10 minutes at 10,000 RPM at four °C; the homogenate is purified through a 0.20 µm syringe filter and injected into GC-FID. All samples were collected in the same manner as described above.

Analytical method validation

System suitability, linearity, and range

System suitability testing is a critical aspect of GC analysis that ensures the reliability and performance of the GC system before sample analysis. It involves evaluating and confirming the suitability of the system by assessing its various parameters and criteria. System suitability testing ensures the comparability of data generated from different samples or batches analyzed using the same GC system.

Several parameters were used to assess the system's suitability to confirm the instrument's performance as well as the reliability of analysis results for the different samples. The retention time of specific compounds or reference standards is analyzed to ensure their consistent retention time across multiple injections. CPF retention time was 13.3 min with RSD of only ±0.35. The shape of chromatographic peaks is also assessed to ensure they are symmetrical, Gaussian, and free from tailing, fronting, or splitting. Distorted peak shapes can indicate issues such as column degradation, contamination, or injection problems. The tailing factor was calculated to be 1.09 ± 0.10 (RSD). The column efficiency is evaluated by measuring the theoretical plates or the plate number of a known compound. The number of theoretical plates (N) indicates the separation efficiency of the column. The obtained results of N showed a reasonably good figure of about 45206 ± 2.95 (RSD).

Linearity is an essential parameter in the validation of GC methods. It assesses the relationship between the analyte concentration and the corresponding detector response, ensuring that the method produces accurate and reliable results over a specified concentration range. CPF peak area was used to plot the calibration curve across the concentration range of 20.0-100.0 µg/mL (Refer to Supplementary materials Figure S3A). The calibration curve generated from the linearity study is subjected to regression analysis to determine the linearity of the response where it was calculated to be (R^2 = 0.9909). The linear equation obtained was found as (y= 2048.7x—20488).

Accuracy, precision, and uncertainty

Accuracy refers to how closely the calculated concentration matches the actual concentration, whereas precision

measures the consistency of the results obtained from a series of measurements. This research conducted a repeatability test, where multiple measurements were made during the same run and day, and an intermediate precision test by comparing the results obtained on different days.

The findings revealed that the average recovery percentage was 99.88%, and the %RSD was 1.50 and 1.38 for repeatability and inter-day precision. An Analysis of Variance (ANOVA) conducted between repeatability and inter-day precision sets reveals p-values of 0.09, 0.81, and 0.99 for 20.00, 60.00, and 100 µg/mL, respectively. These results indicate their differences are not statistically significant and affirm the method's reliability [37]. These results fell within the accepted limits of recoveries between 98 and 102% and %RSD less than 2%, indicating the closeness of calculated to true values and reproducibility of results according to the recommended acceptable criteria set by the United States Food and Drug Administration guidelines [38]. The 95% uncertainty CI for concentrations 40.00, 60.00, and 80.00 µg/mL were 2.64, 2.86, and 1.42, respectively.

Limit of detection (LOD) and limit of quantification (LOQ)

LOD, used for screening purposes, refers to the minimum amount of an analyte present in a sample that can be detected but not necessarily quantified under established experimental conditions. LOQ is the smallest analyte in a sample that can be quantified accurately and precisely under specified experimental conditions. In our method, LOD was 0.29 µg/mL. Meanwhile, LOQ was 0.87µg/mL. Table 2 summarizes the previous analytical validation parameters.

Robustness and ruggedness

Robustness and ruggedness are two important concepts in GC analysis that evaluate the reliability and reproducibility of the method under varying experimental conditions. The purpose of assessing robustness is to evaluate the method's tolerance to minor variations and to determine the critical parameters that significantly influence the results. The robustness of an analytical method is its ability to withstand minor, deliberate variations in analytical parameters. This indicates the reliability of the method when used regularly. Meanwhile, the ruggedness evaluates the method's reproducibility and reliability when performed by different analysts, on different instruments, or in different laboratories. The robustness was calculated by minor changes in the injector's temperature (±5 °C) and detector's temperature (±2 °C) and then calculating the %RSD of results after changes. Ruggedness was calculated the same way. After changing the operating analyst on the same sample over two consecutive days, it was found to be less than 2%, indicating that the method is effective and stable, as shown in Table 3. The

Table 2 Analytical Method Validation

Analytical method validation																		
Range	Equation	R^2	Accuracy (Recovery%)			Precision (%RSD)							System Suitability			LOD	LOQ	
						20	60	100					Tailing	N				
20–100 µg/mL	y = 2048.7x—20488	0.9909	40	60	80	Repeatability			Intermediate Precision				1.09	45206		0.29	0.9	
			99.71	100.84	99.09	1.31	1.47		1.73	1.47								
											0.75	1.92						

Table 3 Robustness and Ruggedness

	Robustness				Ruggedness
	Injector temperature (195 °C)	Injector temperature (205 °C)	Detector temperature (308 °C)	Detector temperature (312 °C)	Change analyst over two days
%RSD	0.975	1.543	1.488	0.797	1.205

data obtained for ruggedness are shown in Supplementary Materials Table S4.

Bioanalytical method validation

Linearity, range, and selectivity

The calibration curve was constructed using QC samples prepared in the plasma over the 20.0-100.0 µg/mL range (Refer to Supplementary materials Figure S3B). The regression equation was calculated as (y= 1494.8x—15456), and the coefficient of determination was (R^2= 0.9905).

A selective bioanalytical method can distinguish and quantify an analyte in a biological matrix, even if other interfering substances are present. The Acceptance criteria demand that interfering components, if present, should not exceed 20% of LLOQ. The six plasma loads that were tested did not have any significant interference at CPF retention time.

Accuracy, precision, uncertainty, and recovery (extraction efficiency)

Once linearity, range, and selectivity have been established, the bioanalytical method's accuracy and precision must be determined. To evaluate accuracy and precision, the QC samples should be within ±15% except for LLOQ, which can be within ±20%. The bioanalytical method's average accuracy and precision was 97.1% ±1.4. The 95% uncertainty CI for LLQC, LQC, MQC, and HQC were 0.71, 1.32, 1.35, and 2.12, respectively.

In recovery studies, no defined acceptance limits are set to assess extraction efficiency. Nonetheless, the extraction recoveries should be consistent and reproducible. The average extraction recoveries were, as recommended, around 71.96%. Table 4 summarizes the bioanalytical validation parameters obtained.

Sensitivity, matrix effect, and carryover

After injecting extracted LLOQ six times, the average %RSD was 1.73, showing the method's ability to detect low CPF concentration with reproducible results.

The term "Matrix effect" describes any changes in the accuracy or precision of analysis due to unintended substances in the sample matrix, which may interfere with the analysis and cause errors. According to ICH guidelines, the acceptance criteria is that LQCs' and HQCs' accuracy and precision are to be within ±15% of the bio-analytical validation results when assessed in different plasma batches. The three replicates of these QCs spiked in six different plasma batches were evaluated, and the results didn't exceed the stated limit.

Carryover is the appearance of an analyte signal in a sample after analysis of another preceding sample. Carryover is assessed by injecting a blank sample immediately after a high-concentration sample. The absence of significant peak signals in the blank sample indicates the absence of carryover, ensuring the integrity of subsequent analyses. The acceptance criteria for carryover are that the response, if present, shouldn't exceed 20% of the LLOQ. Blank samples injected after HQC runs were within the limit.

Stability studies

Stability studies assess every stage in sample handling, including preparation, processing, analysis, transport, and storage, to ensure that the concentrations of samples remain unaffected. The mean recoveries for bench-top

Table 4 Bio-Analytical Method Validation

Bio-Analytical Method Validation							
Range	Linearity Equation	R^2	Standard (Concentration)	LQC (40 µg/mL)	MQC (60 µg/mL)	HQC (80 µg/mL)	Sensitivity LLQC (20 µg/mL)
20–100 µg/mL	y = 1494.8x—15456	0.9905	Accuracy (R%)	94.51	99.34	97.20	98.94
			Repeatability (%RSD)	1.85	1.01	1.49	1.73
			Intermediate Precision (%RSD)	1.41	1.12	1.32	1.81
			Recovery	74.26	71.68	72.03	69.87

stability were 97.75% for LQC and 97.20% for HQC, 94.57 and 96.10% for freeze–thawing, and 95.17 and 96.13% for long-term stability. These results fall within the ICH acceptance criteria, which state that concentration recoveries must be within ±15%.

Chromatograms and application

The proposed method determined the amount of CPF in commercial products and plasma. Lantrek® was selected as a potential representative sample to showcase the method's effectiveness with a readily available product rather than limiting our findings to a single formulation. The experiment was conducted six times, and a satisfactory recovery result of 99.28 ±1.63 was obtained, which agrees with the label claims. Figure 2 and Supplementary Figure S5 show the blank and the application chromatograms.

Limitations, and future perspectives

The use of CPF, OP pesticide, presents several challenges and concerns over its potential health effects, particularly neurological impacts on humans and wildlife. Although being restricted/banned in several countries, it's still being used in massive amounts worldwide and its persistence in the environment raises issues related to soil and water contamination. Until safer and more sustainable pest management practices are introduced, the analysis of CPF should become more effective and relevant in environmental and food safety contexts. The most important factor is detection limits where traditional methods may struggle with low concentrations, and matrix effects that can interfere with accurate detection due to complex sample compositions. The fast degradation of CPF in environmental samples can also complicate its analysis. Chromatographic methods present enhanced sensitivity and selectivity for CPF analysis. However, real-time monitoring through sensors and portable devices may provide on-site analysis. Efforts should be made

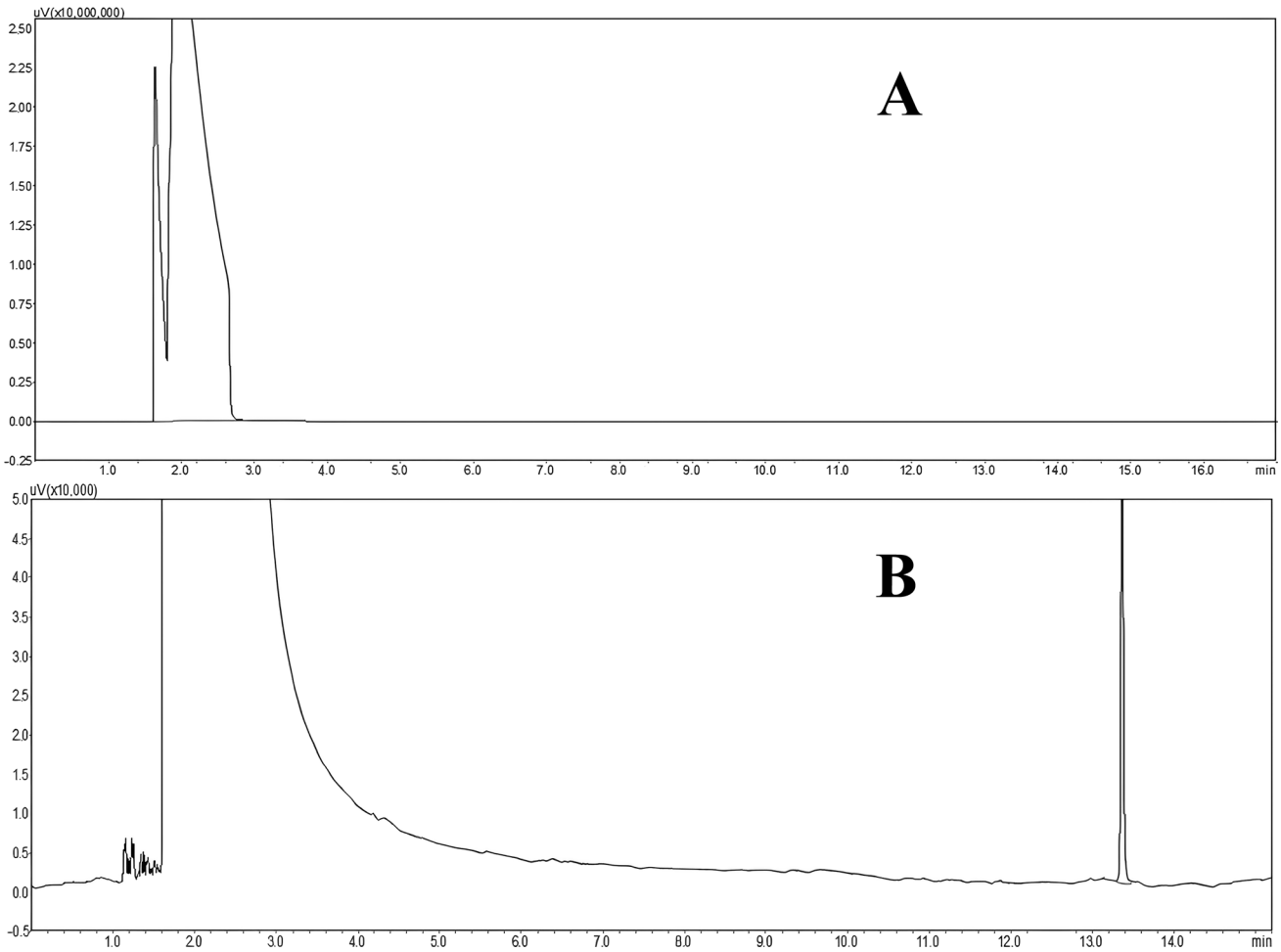


Fig. 2 Chromatograms showing blank plasma (A) and CPF standard in spiked Plasma (100 µg/mL)



Fig. 3 Greenness assessment of the proposed method to a reported GC–MS method [41], both applied for CPF determination in plasma, according to the AGREE metric

toward standardizing the analytical methods and global regulations to facilitate comparability and compliance.

Greenness assessment

Green chemistry has become a primary goal for encouraging sustainable growth and developing eco-friendly materials and processes. Based on 12 principles, green analytical chemistry (GAC) considers health, environmental, and safety issues during the analysis, and many metrics have been developed for its assessment [39]. Assessing the ecological impact of analytical methodologies is of significant importance due to the potential environmental consequences associated with the use of such techniques. Assessing the ecological impact allows for understanding the environmental footprint of these consumables, including their production, usage, and disposal [40]. GC analysis also can generate chemical waste, such as spent solvents, sample residues, and column maintenance materials. Evaluating the ecological impact allows for developing appropriate waste management strategies, including recycling, reuse, or safe disposal methods. By assessing the ecological impact of GC analytical methodologies, researchers, laboratories, and industries can minimize their environmental footprint, reduce pollution, and contribute to sustainable practices.

One of the most reliable metrics, Analytical Greenness (AGREE), is used to evaluate and compare the proposed method to a GC-MS method [41], both applied for CPF determination in plasma. The AGREE metric output is represented in a clock-like format with an overall score and color representation in the center of the pictogram. The closer the score to 1, the greener the assessed procedure. Additionally, AGREE is based on assessing the performance of the procedure across each of the 12 principles of green chemistry introduced by Anastas and Warner [42, 43]. The color code is reflected with a red-yellow-green scale, where green indicates an ecologically safe procedure, while red represents the lowest safety in their corresponding segment [44, 45]. The proposed method scored 0.63 compared to the reported GC-MS method, which scored 0.52. This score difference is mainly due to shorter analysis time (Segment 8), FID's lower energy consumption (Segment 9), and the use of milder, fewer solvents (Segment 11 and 12) and steps (Segment 4) during sample preparation, as shown in Figure 3.

Conclusion

A new method for detecting Chlorpyrifos in pesticide products and human plasma has been developed and validated. The technique uses rapid, cost-effective, sensitive, and precise gas chromatography with flame ionization detection (GC-FID). The approach requires no derivatization, has a straightforward sample preparation, and uses minimal solvents, making it eco-friendly. The method can be used for routine quality control analysis of Chlorpyrifos given the increased scrutiny and regulation Chlorpyrifos has recently faced, which may lead to its screening in various products and biological samples in the future as a banned or counterfeit pesticide.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00769-025-01643-z.

Author contributions Conceptualization contributed by K.G. and A.I.; methodology, software, and validation contributed by A.I., S.A., G.A., and B.S.; formal analysis contributed by K.G., A.I., and B.S.; investigation contributed by K.G. and A.I.; data curation contributed by K.G., S.A. and A.I.; writing—review and editing contributed by A.I., S.A., and B.S.; resources and project administration contributed by S.E. and A. A.

Funding This research received no external funding.

Data availability Data are provided within the manuscript or supplementary information files.

Declarations

Conflicts of interest The authors declare no competing interests.

Ethics approval Not applicable.

Informed Consent Not applicable

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